



1. The bubble sort algorithm is a simple sorting algorithm that goes through the list, compares adjacent elements and swaps the elements if they are in wrong order. This process is repeated N number of times, where N is the number of elements in the list, until the list is sorted. The C code for Bubble sort algorithm is given below. Convert below given C code to assembly code. **(10)**

```
#define SIZEC 10
//Each number is always smaller than 64
int c[SIZEC]={23, 12, 5, 44, 98, 53, 6, 89, 32, 65};
void main ()
{
    //printf("Array before bubble sort\n");
    //for(int j=0; j<SIZEC; j++) printf("%d,", c[j]);
    //printf("\n");

    int swapped;
    do
    {
        swapped = false;
        for (int i =1; i< SIZEC; i++)
        {
            if(c[i-1] > c[i])
            {
                int temp = c[i];
                c[i]=c[i-1];
                c[i-1] = temp;
                swapped = true;
            }
        }
    } while (swapped);

    //printf("Array after bubble sort\n");
    //for(int j=0; j<SIZEC; j++) printf("%d,", c[j]);
    //printf("\n");
    return 0;
}
```